

Ex.no:9	MINI PROJECT – HEALTHCARE ANALYTICS

Domain and Problem Statement

Domain: Healthcare Data Analytics & Hospital Performance Analysis

Problem Statement:

Healthcare organizations generate large volumes of hospital-related data but often lack clear insights into hospital distribution, ratings, reviews, and regional patterns. This project addresses these challenges by transforming hospital data into structured business intelligence using Power BI and Tableau, enabling analysis of hospital distribution across states and cities, average ratings, review patterns, and overall hospital performance to support data-driven healthcare analysis.

Dataset Description

Dataset Name: Hospital Analytics Dataset

Granularity: Each row represents a single hospital.

Key Entities & Attributes:

- **Hospital & Location:** id, City, State, District
- **Geographical Information:** Density, Latitude, Longitude
- **Hospital Performance:** Rating, Number of Reviews

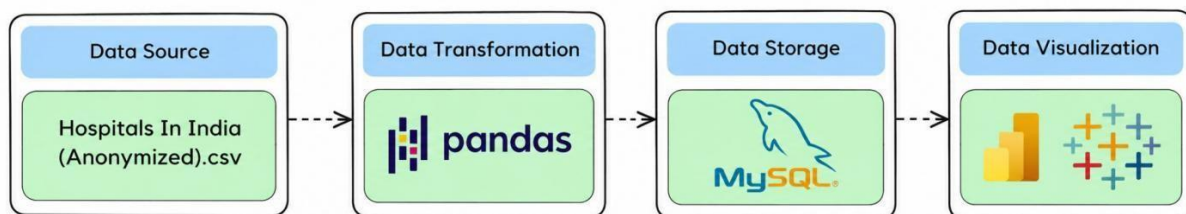
Data Source:

Primary Source: Hospital Analytics Dataset prepared for healthcare and hospital performance analysis.

Storage Infrastructure:

Cleaned and pre-processed hospital records maintained in a CSV dataset and imported into Microsoft SQL Server, Power BI and Tableau for interactive dashboard development and analysis.

Architecture:



Program:

1. Python – Data Cleaning & Transformation

```
import pandas as pd

df = pd.read_csv("Hospitals In India (Anonymized).csv")

hospitals = df.copy()

hospitals["City"] = hospitals["City"].fillna("Unknown")
hospitals["State"] = hospitals["State"].fillna("Unknown")
hospitals["District"] = hospitals["District"].fillna("Unknown")

numeric_columns = [
    "Density",
    "Latitude",
    "Longitude",
    "Rating",
    "Number of Reviews"
]

for column in numeric_columns:
    hospitals[column] = hospitals[column].fillna(0)

hospitals.drop_duplicates(inplace=True)

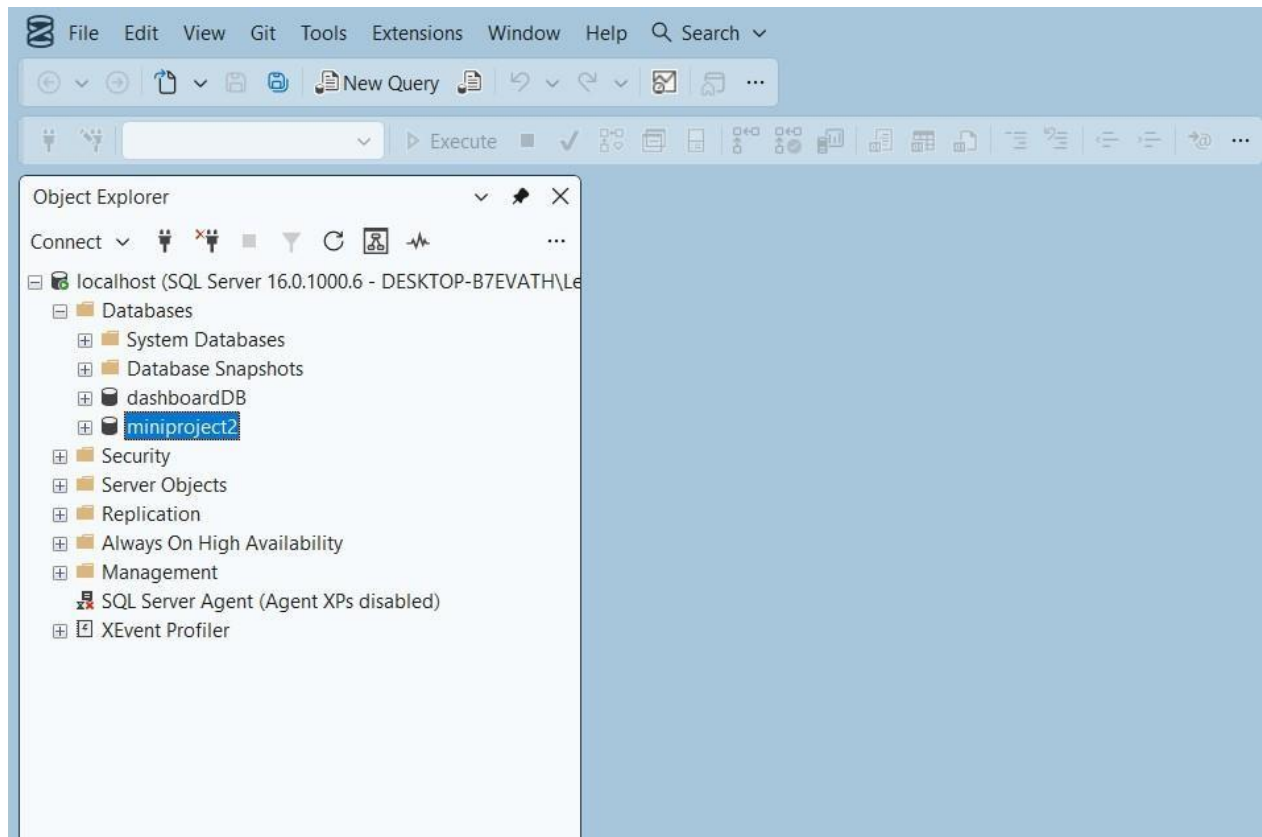
hospitals.columns = (
    hospitals.columns
    .str.lower()
    .str.strip()
    .str.replace(" ", "_"))

print("Original Shape:", df.shape)
print("Cleaned Shape:", hospitals.shape)
print("\nDataset Preview:")
print(hospitals.head())
print("\nDataset Information:")
print(hospitals.info())
```

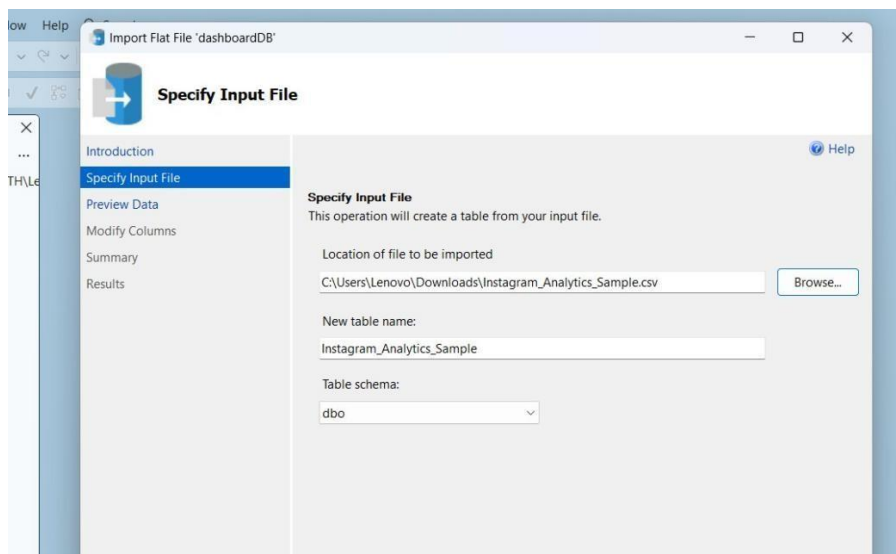
2. SQL SERVER CONNECTIVITY

INSTALL SQL SERVER MANAGEMENT STUDIO

1) Create database named “miniproject2”

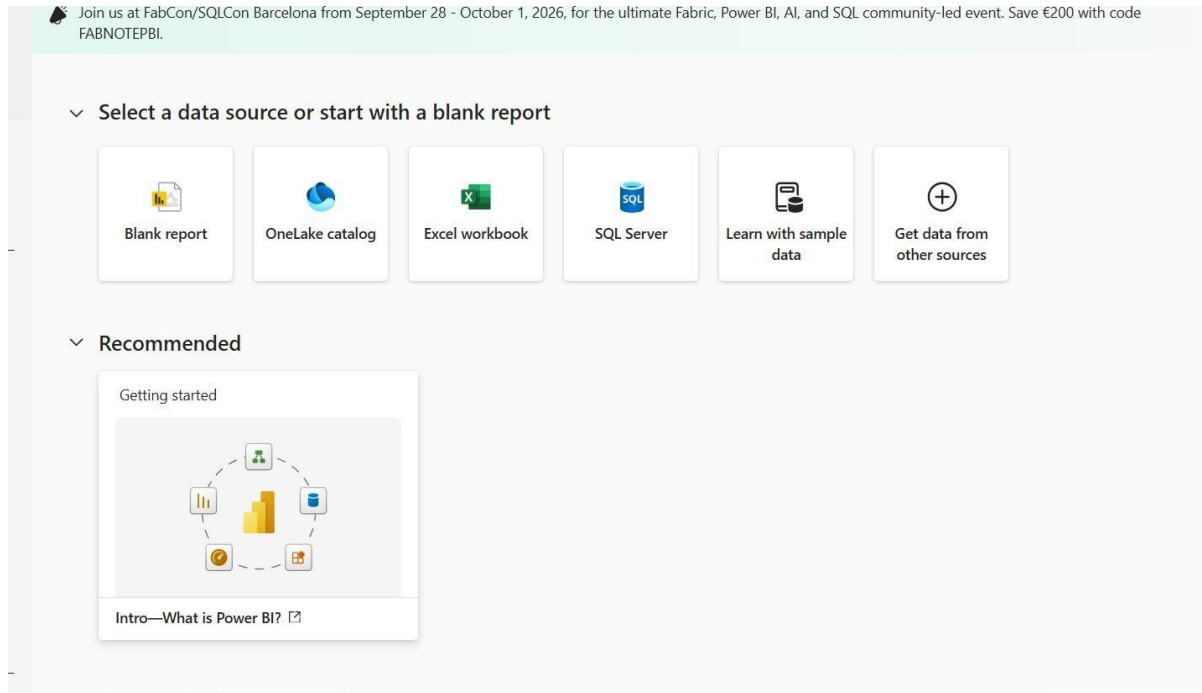


2) Import the csv file of the dataset

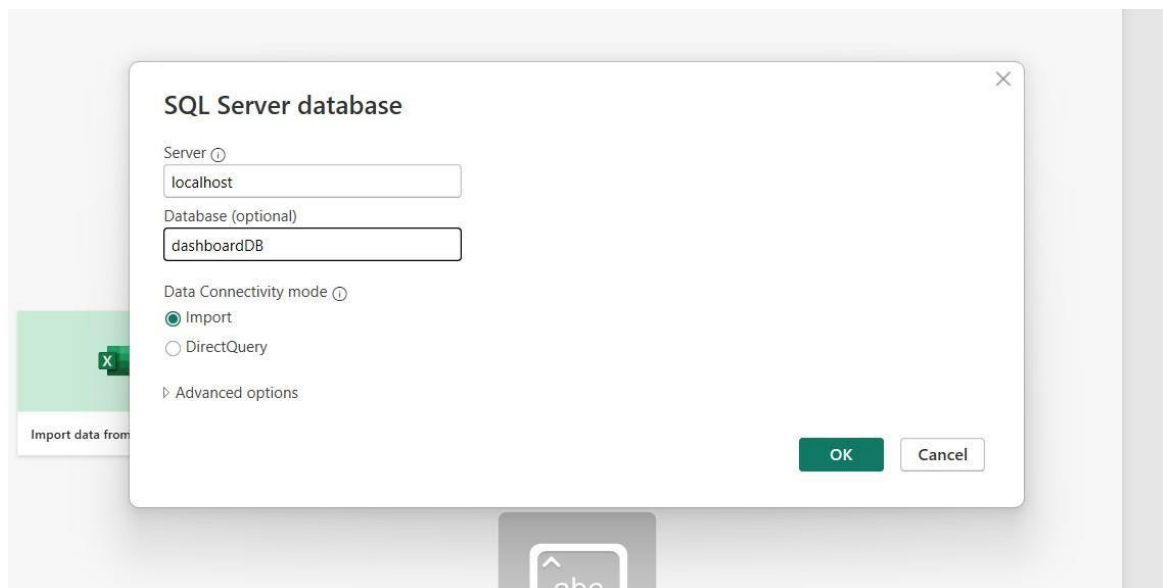


POWER BI

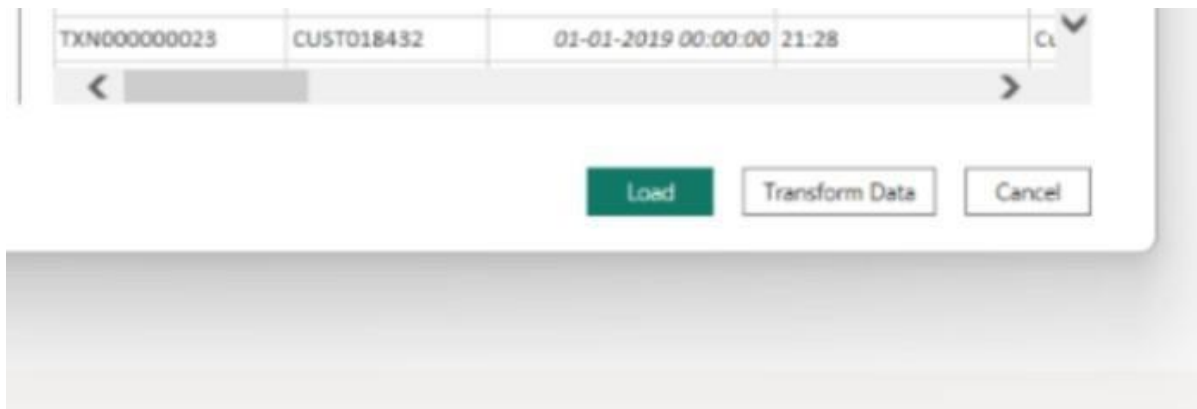
3) Now go to MICROSOFT POWER BI



4) Click SQL SERVER



5) Click OK and click transform data



6) Create DAX measures

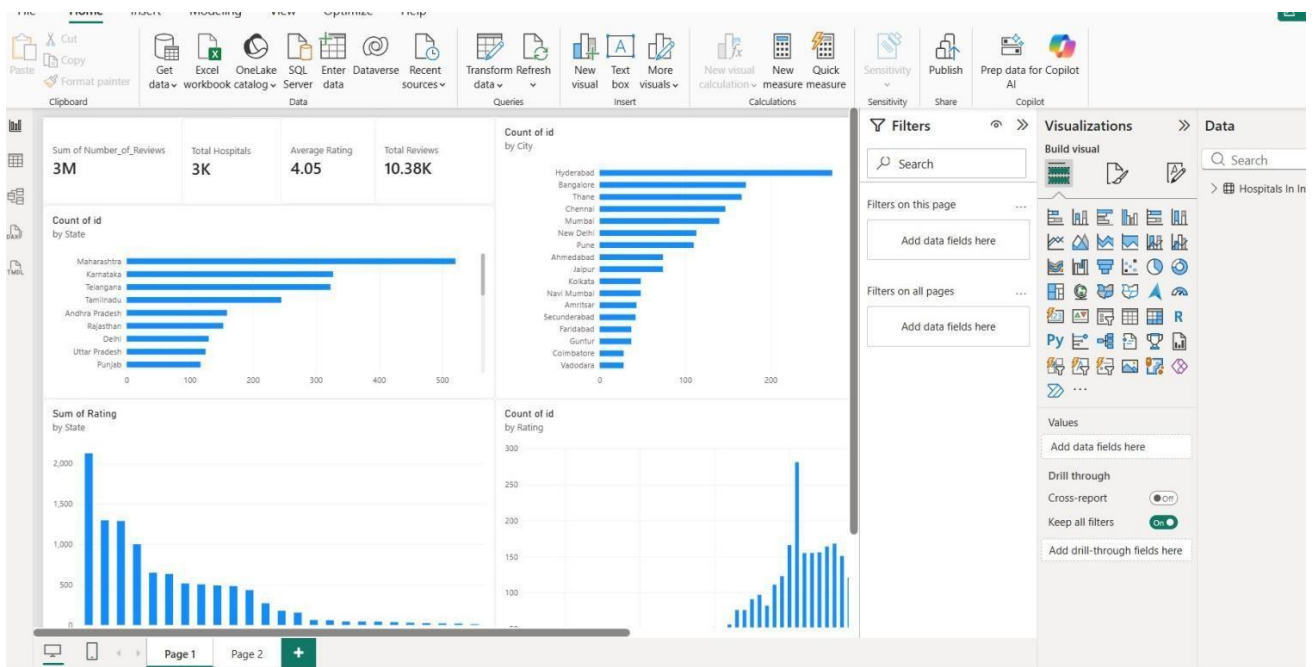
TOTAL HOSPITALS :

```
1 Total Hospitals = COUNT('Hospitals In India (Anonymized)'[id])
```

AVERAGE RATING :

```
1 Average Rating = AVERAGE('Hospitals In India (Anonymized)'[rating])
```

7) Power BI Final Executive Dashboard & Drill-Through

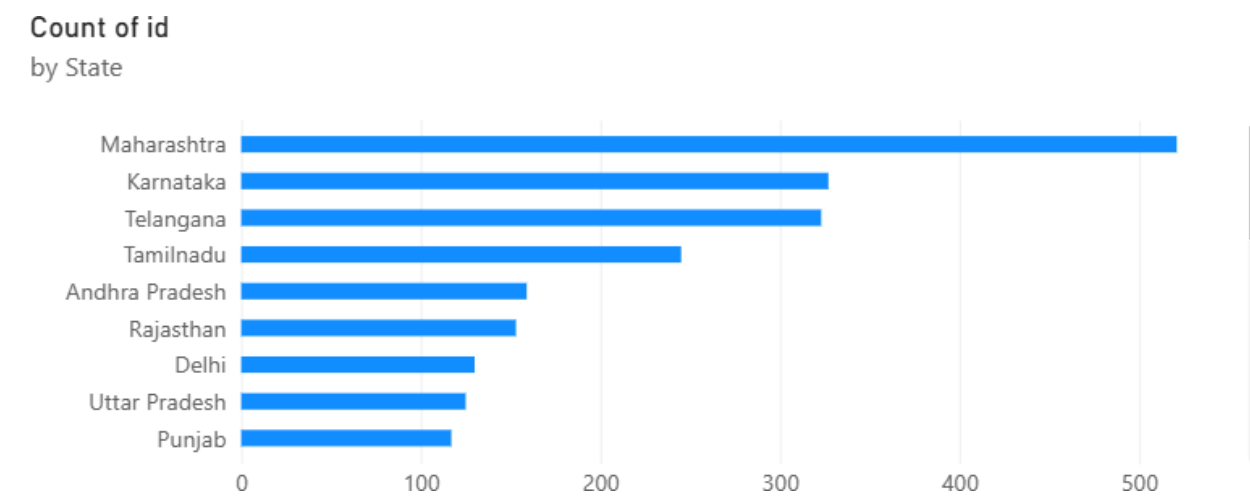


1. Executive Canvas:

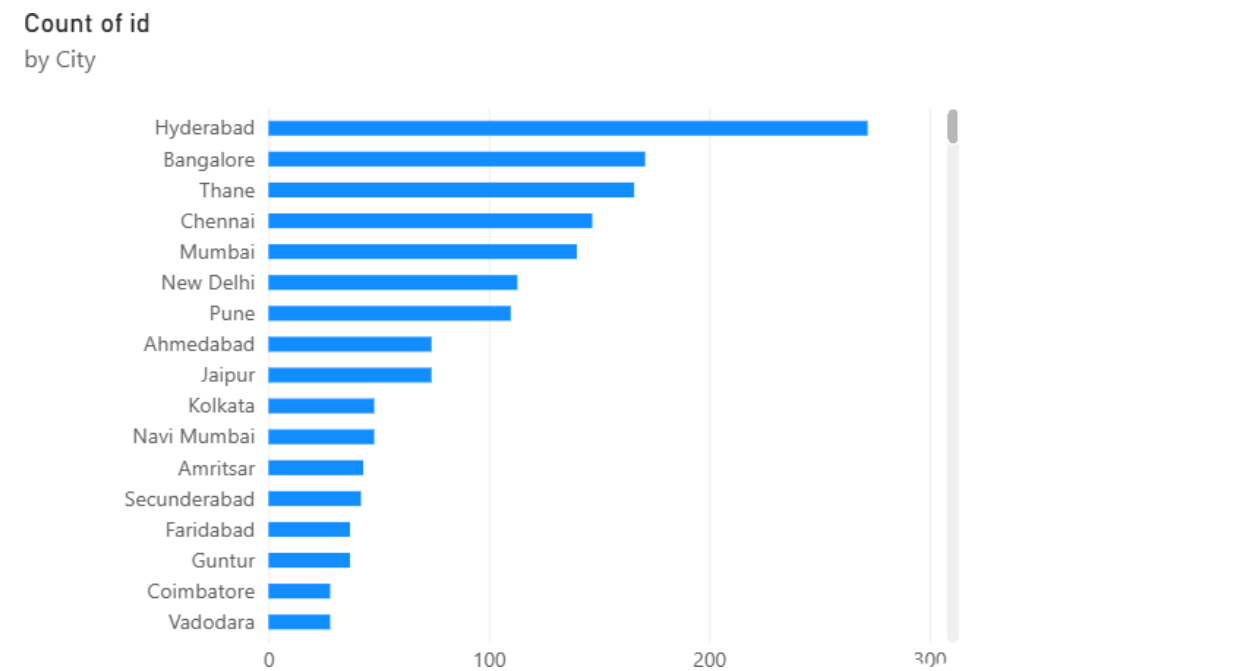
Place Card Visuals across the top:



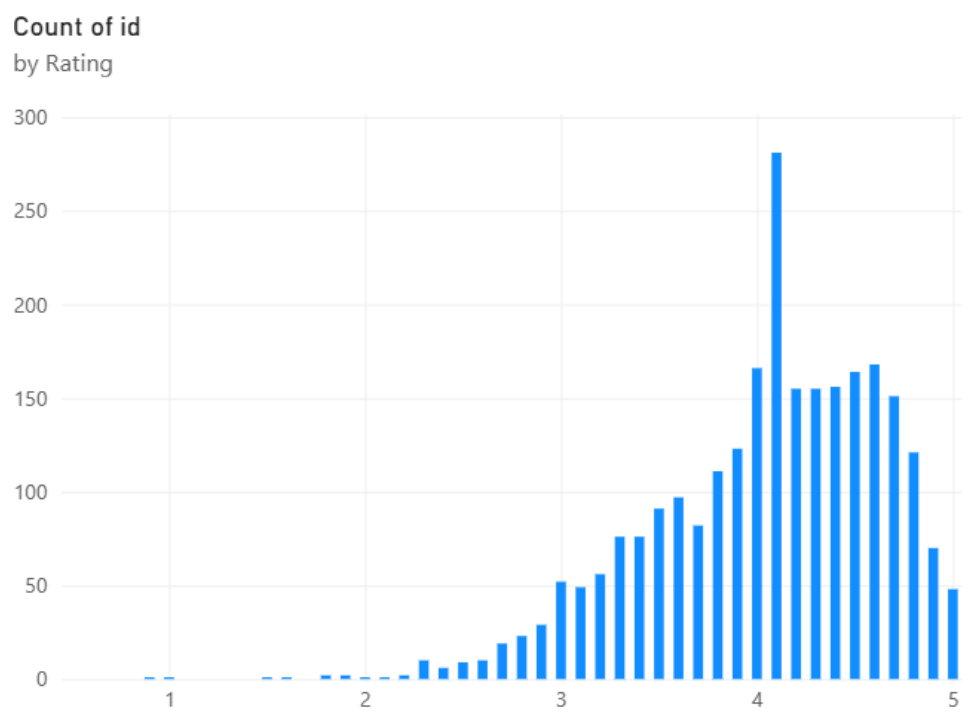
Visualise number of hospitals in each state



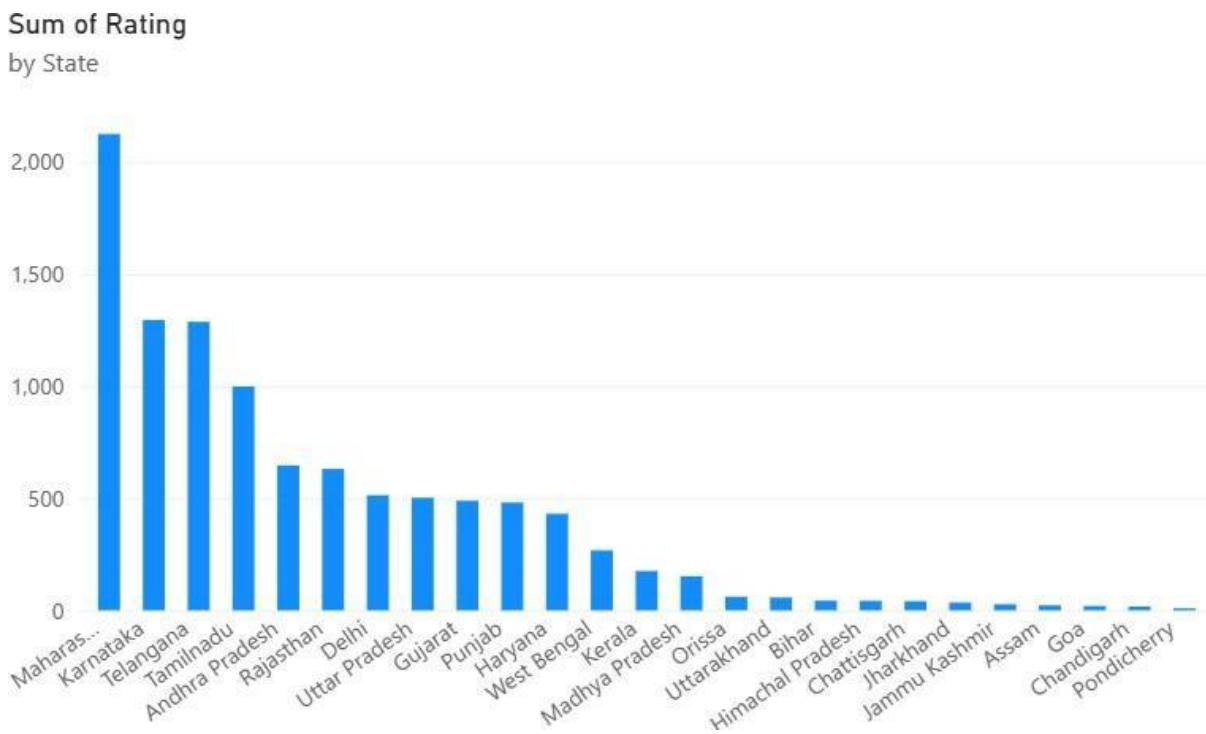
count number of hospitals by city



Count of id by rating



Total rating by states



Create 2 slicers by state and city

State	City
☐ Andhra Pradesh	☐ Agra
☐ Assam	☐ Ahmedabad
☐ Bihar	☐ Ahmednagar
☐ Chandigarh	☐ Ajmer
☐ Chattisgarh	☐ Aligarh
☐ Delhi	☐ Allahabad
☐ Goa	☐ Aluva
☐ Gujarat	☐ Alwar
☐ Haryana	☐ Ambala
☐ Himachal Pradesh	☐ Amritsar
☐ Jammu Kashmir	☐ Anantpur
☐ Jharkhand	☐ Aurangabad
☐ Karnataka	☐ Bahadurgarh
☐ Kerala	☐ Balotra
☐ Madhya Pradesh	☐ Bangalore
☐ Maharashtra	☐ Banswara
☐ Orissa	☐ Bantwal
☐ Pondicherry	☐ Baramati
☐ Punjab	☐ Bareilly
☐ Rajasthan	☐ Barmer
☐ Tamilnadu	☐ Behrampur
☐ Telangana	☐ Belgaum
☐ Uttar Pradesh	☐ Belguam
☐ Uttarakhand	☐ Bellary
☐ West Bengal	☐ Berhampur

Implement drill through

Values

Add data fields here

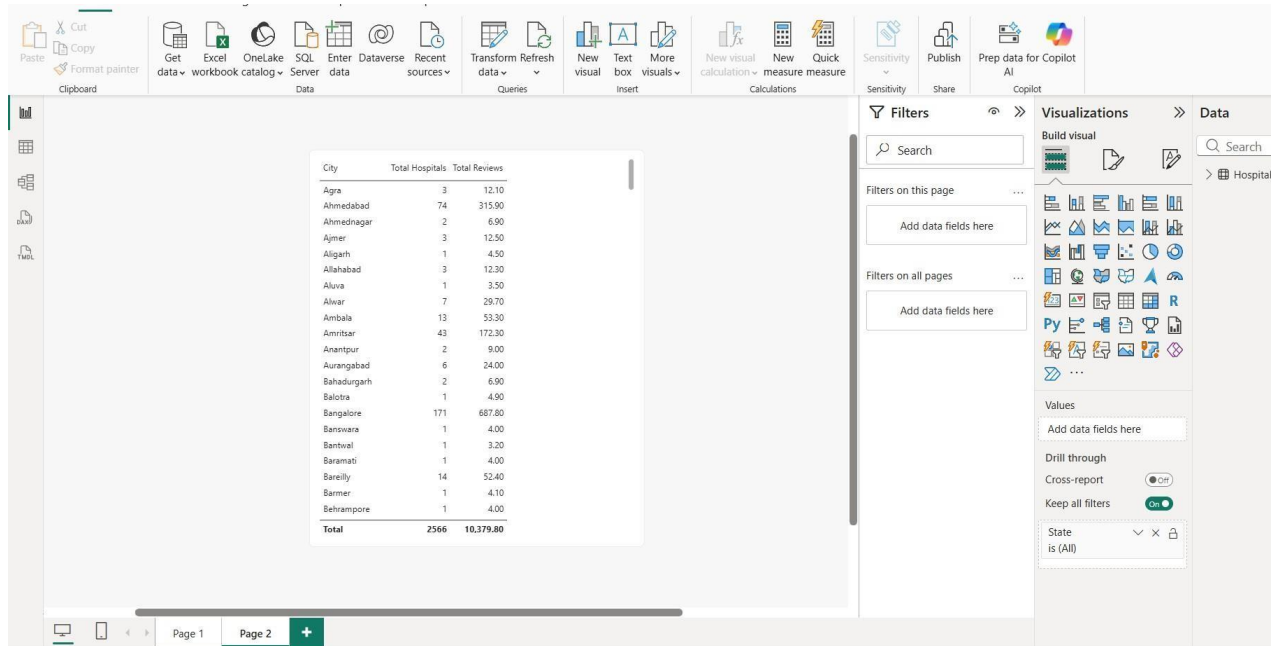
Drill through

Cross-report ☐ Off

Keep all filters ☒ On

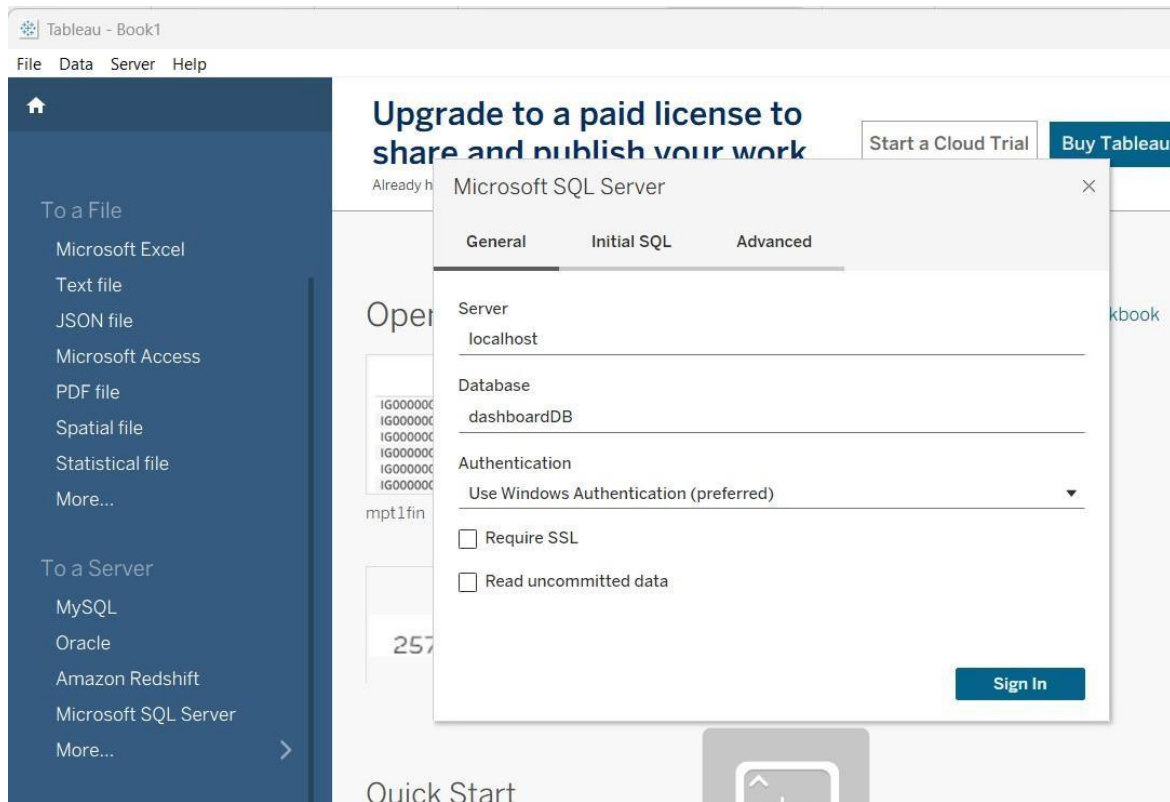
State

is (All)

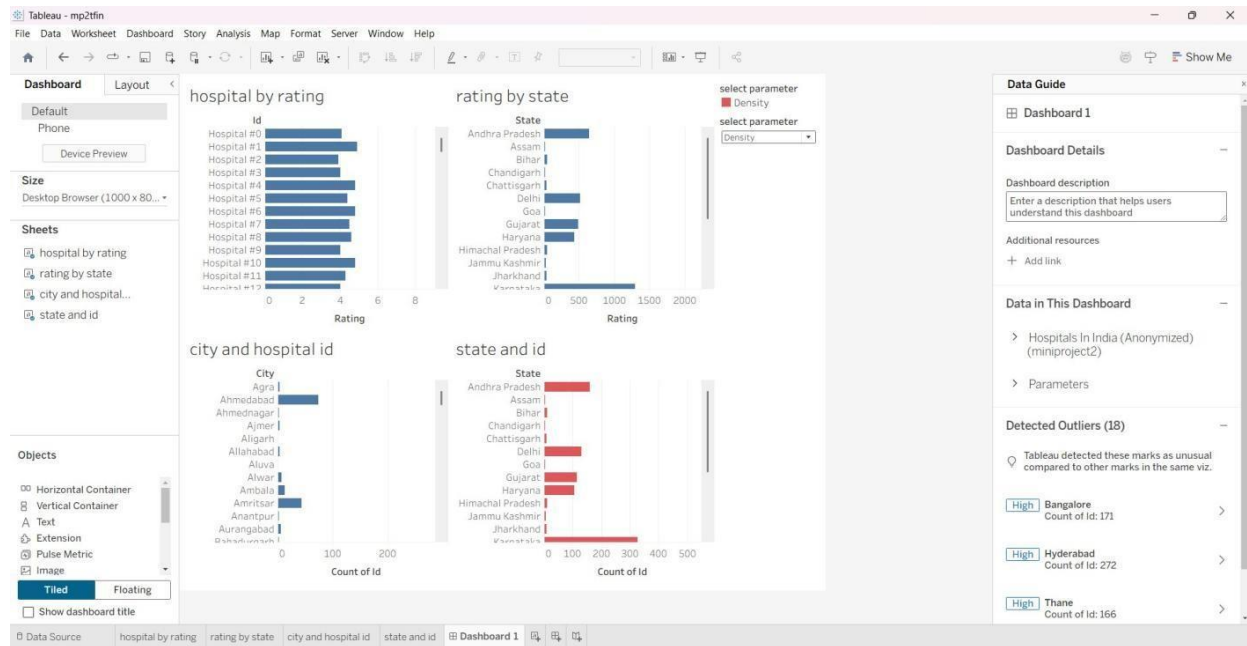


TABLEAU

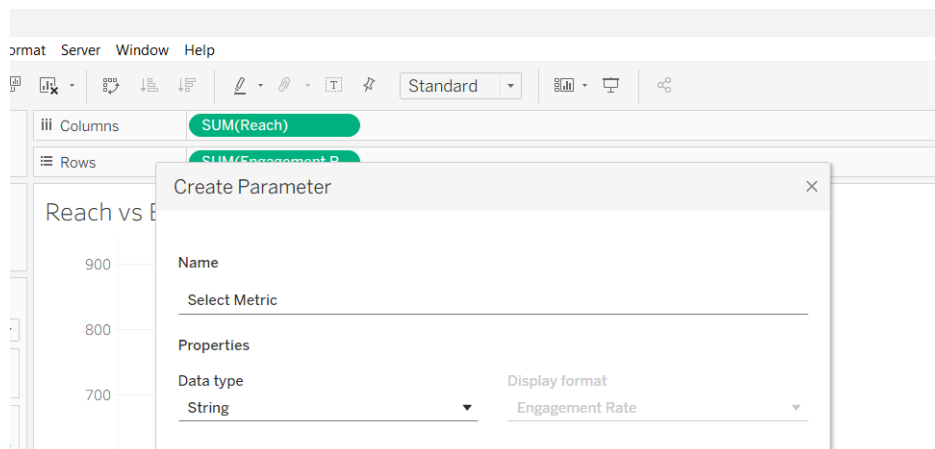
Import the data from sql server



FINAL DASHBOARD :

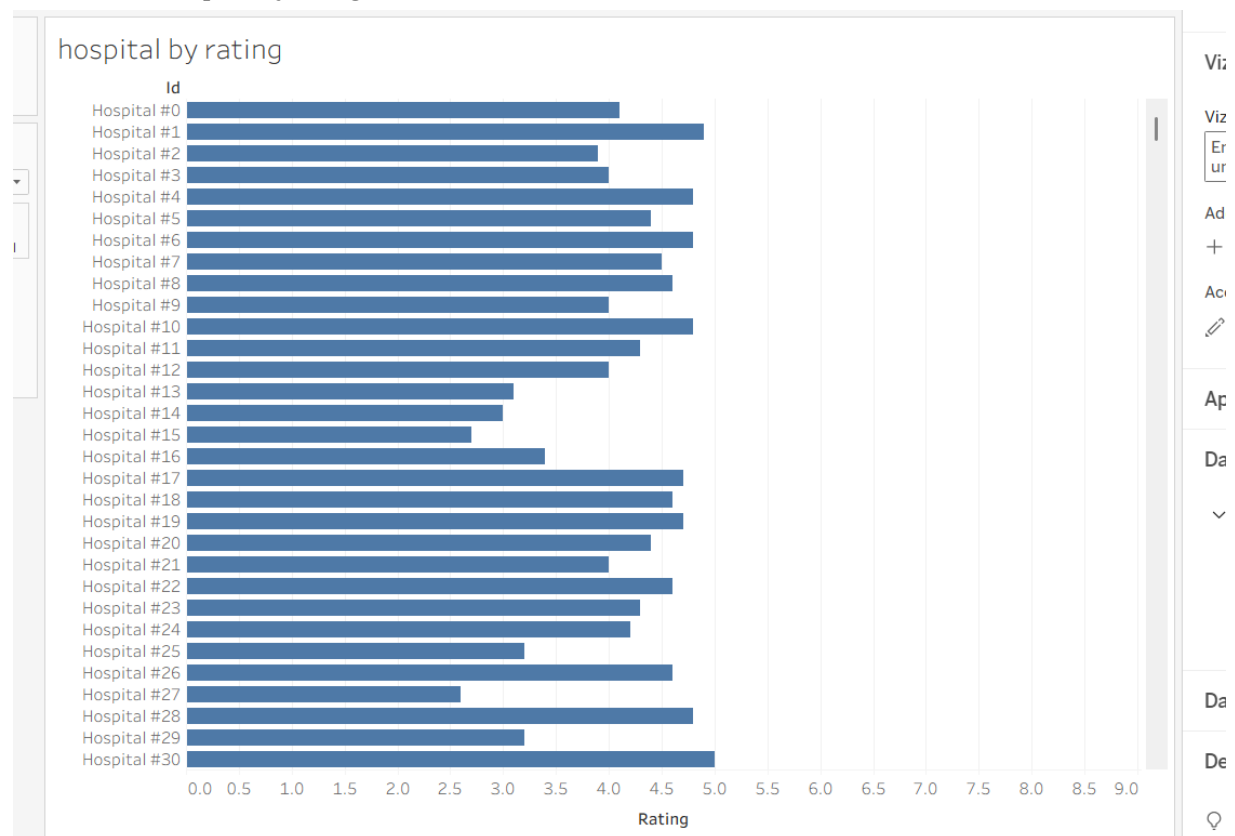


Build parameters

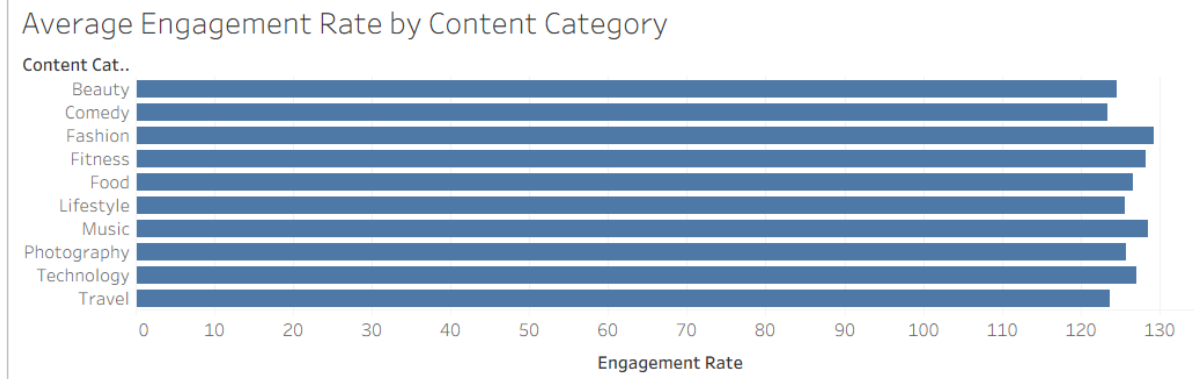




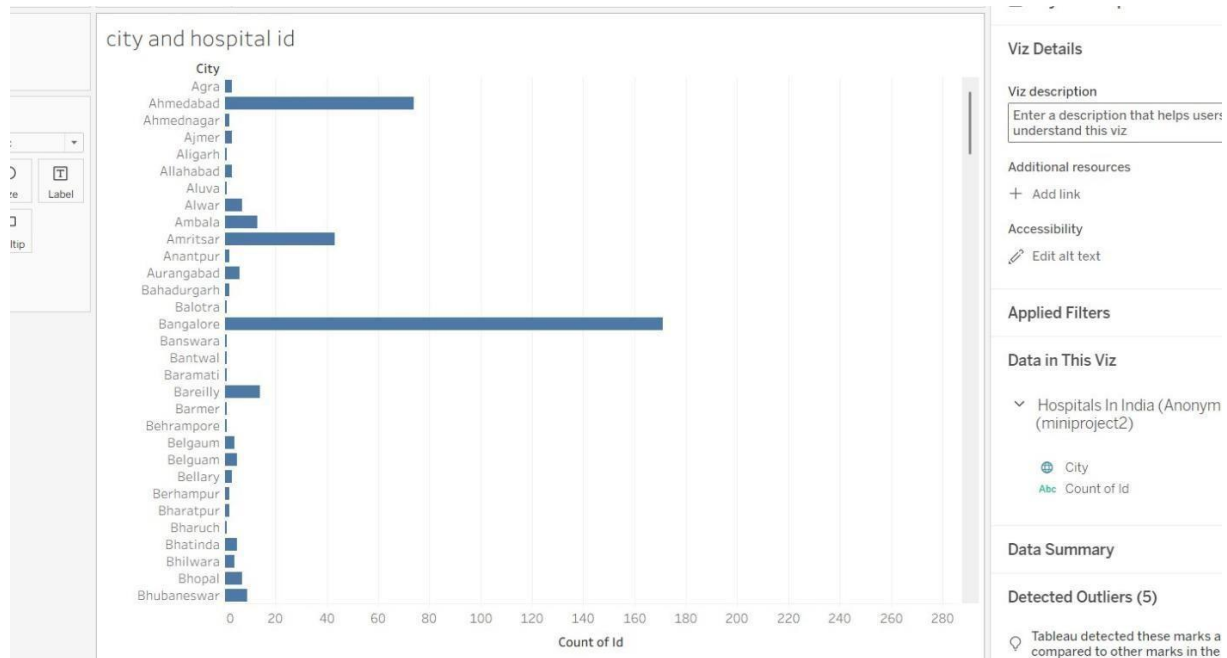
To visualise hospital by rating



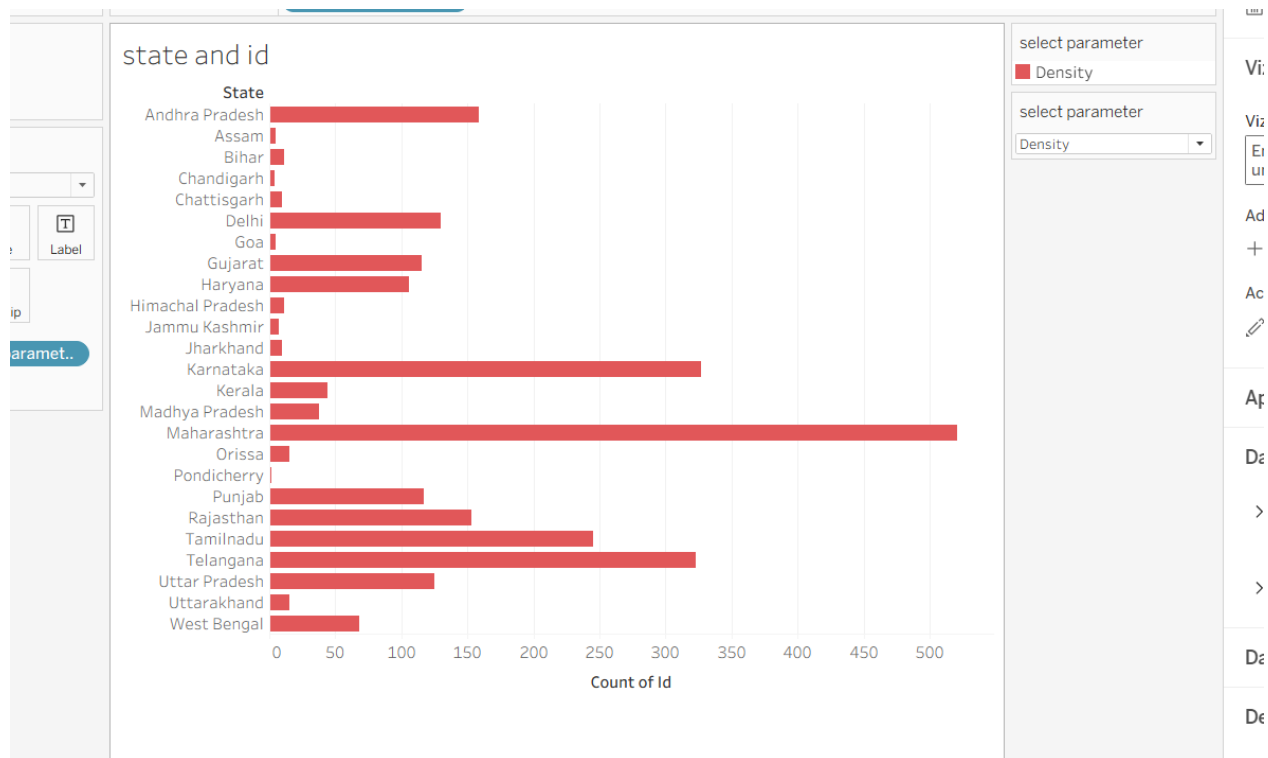
To visualize rating by state



To visualize number of hospitals in each city



To visualise number of hospitals in each state



Observed Insights

1. Hospital Ratings

Most hospitals have ratings between 3 and 5, indicating generally moderate to good ratings. Hospital #1 has the highest rating among the displayed hospitals, close to 5. The ratings are relatively consistent across the hospitals shown in the dashboard.

2. Rating by State

Karnataka has the highest rating-related value among the states displayed, followed by Andhra Pradesh. Delhi, Gujarat, and Haryana also show comparatively higher values than most other states. Several states such as Bihar, Assam, and Chandigarh have very low values in comparison.

3. Hospital Distribution by City

Ahmedabad has the highest hospital count among the cities displayed in the dashboard. Amritsar also has a noticeable number of hospitals compared with most other listed cities. Several cities such as Agra, Ajmer, Aligarh, and Allahabad have relatively fewer hospitals.

4. Hospital Distribution by State

Karnataka has the highest number of hospital IDs, followed by Andhra Pradesh and Delhi. Gujarat and Haryana also contribute a significant number of hospitals to the dataset. States such as Assam, Bihar, Chandigarh, and Chhattisgarh have comparatively fewer hospital